PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE B · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

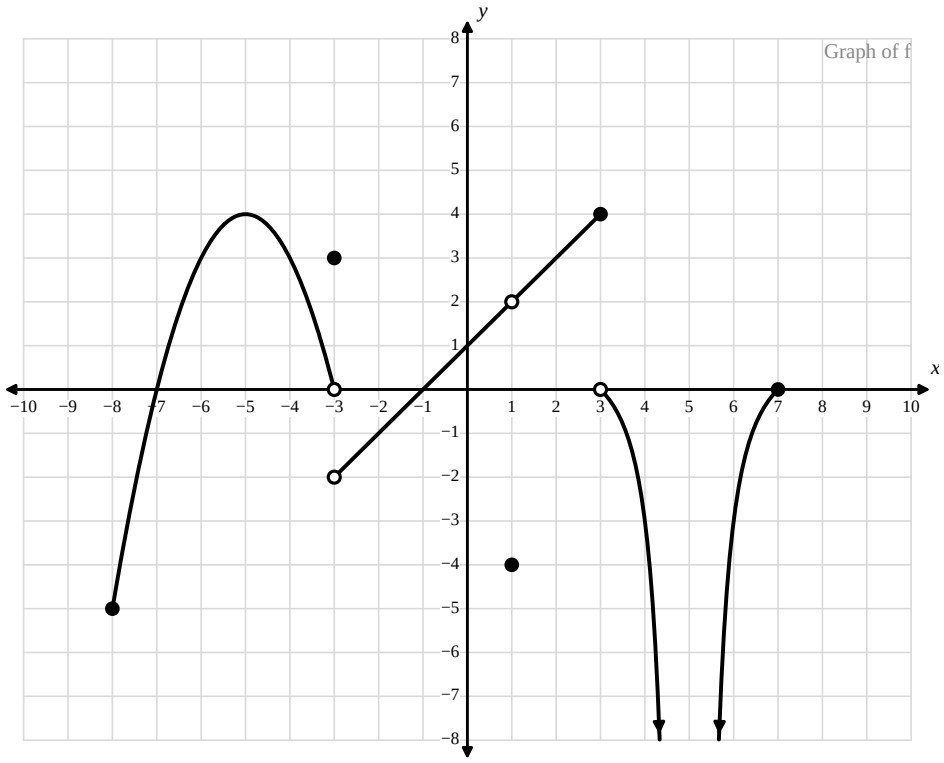
_____

_____

Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

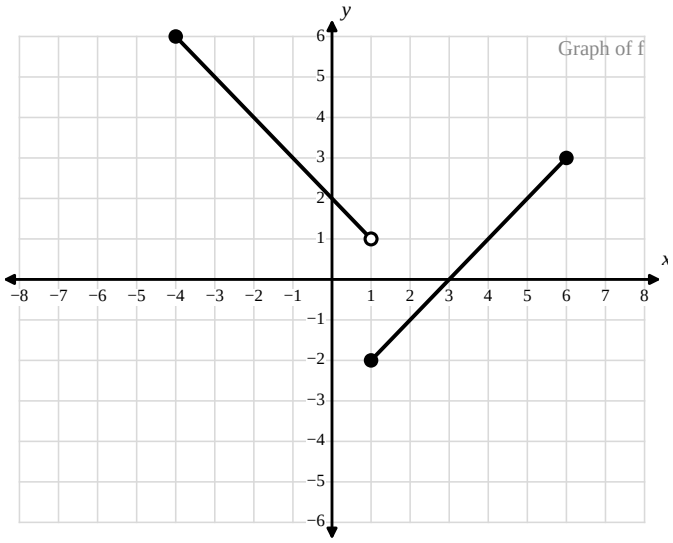


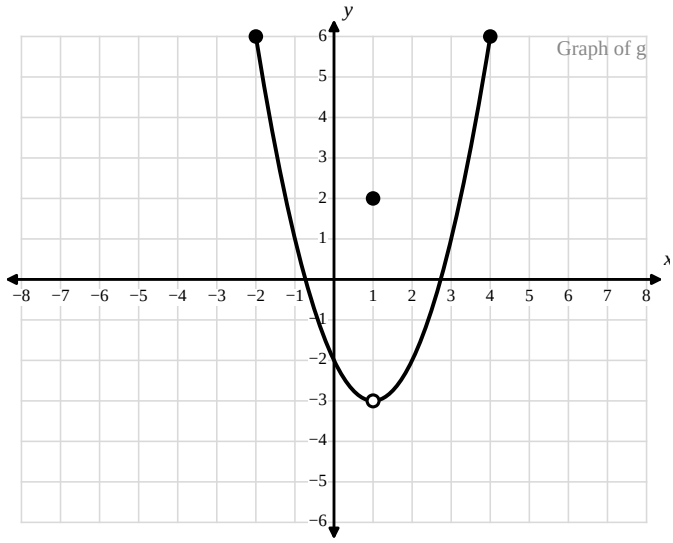
1. $\lim_{x\to 1} f(x) =$	**2.** $f(1) =$	**3.** $\lim_{x\to -3} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 0$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim\limits_{x \to 1^-} \dfrac{2f(x) - g(x)}{f(x)\, g(x)}$

(2 points)

6. Use the table to estimate $\lim\limits_{x \to -1} f(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−1.1	−1.01	−1.001	$\to -1 \leftarrow$	−0.999	−0.99	−0.9
$f(x)$	3.21	3.0201	3.002001		−0.998	−0.98	−0.8

$\lim\limits_{x \to -1^-} f(x) \approx$ _____ $\lim\limits_{x \to -1^+} f(x) \approx$ _____ $\lim\limits_{x \to -1} f(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\displaystyle\lim_{x\to -2^+} \frac{x^2+2x}{|x+2|}$

8. $\displaystyle\lim_{x\to 4} \frac{x^2-16}{x^2-x-12}$

9. $\displaystyle\lim_{x\to 5} \frac{\sqrt{2x-1}-3}{x^2-25}$

10. $\displaystyle\lim_{x\to 1} \frac{\frac{x}{x+1}-\frac{1}{2}}{x-1}$

11. For $f(x)=\sqrt{2x+1}$, find $\displaystyle\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

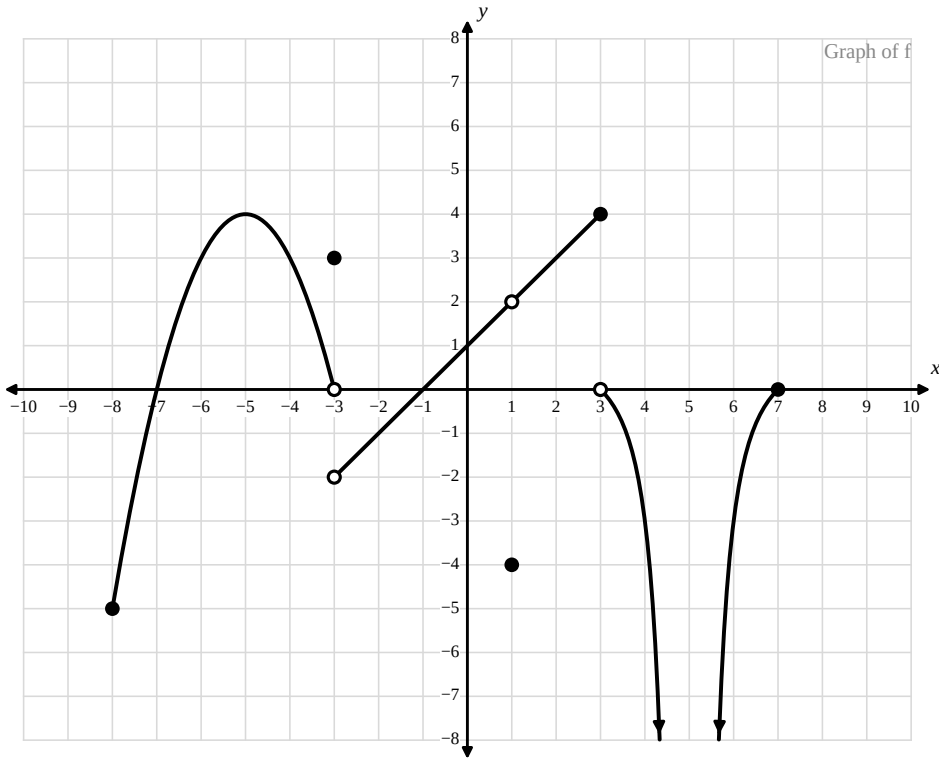
PRACTICE B · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to 1} f(x) =$

Both sides of the line head to the open circle at $(1, 2)$.

2

2. $f(1) =$

The filled dot at $x = 1$ *is at height* -4.

-4

3. $\lim_{x\to -3} f(x) =$

$\lim_{x\to -3^-} f(x) = 0$,
$\lim_{x\to -3^+} f(x) = -2$
$0 \neq -2$, *so* $\lim_{x\to -3} f(x)$ *does not exist.*

DNE

(2 points)

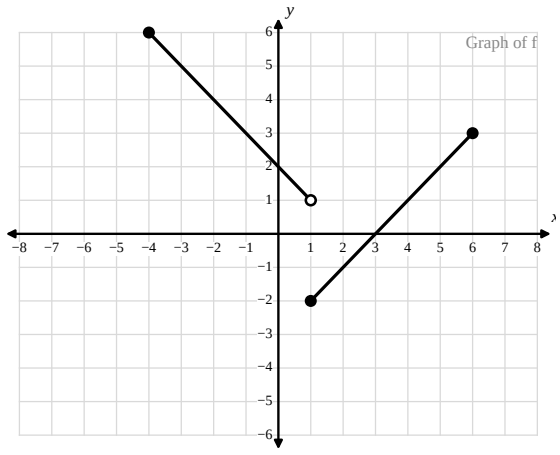
4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 0$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

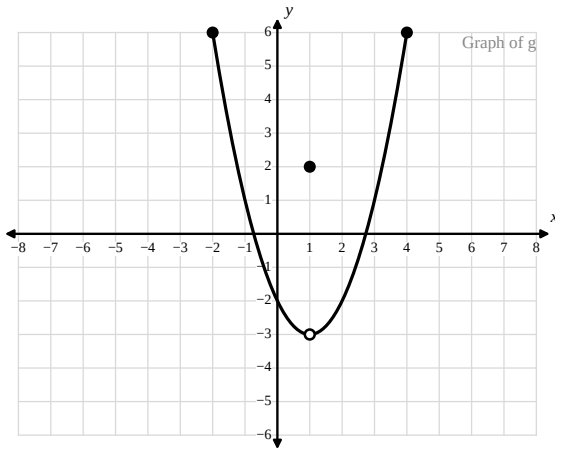
Height 0 is reached from both sides at $x = -7$ *and* $x = -1$.

Not $x = -3$ *(left side only), not* $x = 3$ *(right side only; the left heads to 4), not* $x = 7$ *(graph ends there).*

$a = -7,\ a = -1$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x\to 1^-} \dfrac{2f(x)-g(x)}{f(x)\,g(x)}$

$$\lim_{x\to 1^-} \frac{2f(x)-g(x)}{f(x)g(x)} = \frac{2\lim_{x\to 1^-} f(x) - \lim_{x\to 1^-} g(x)}{[\lim_{x\to 1^-} f(x)]\,[\lim_{x\to 1^-} g(x)]}$$

Quotient (bottom limit is −3, not 0), difference, constant multiple, product.

$$= \frac{2(1)-(-3)}{(1)(-3)} = \frac{5}{-3} = -\frac{5}{3}$$

$-\frac{5}{3}$

(2 points)

6. Use the table to estimate $\lim_{x\to -1} f(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−1.1	−1.01	−1.001	→ −1 ←	−0.999	−0.99	−0.9
$f(x)$	3.21	3.0201	3.002001		−0.998	−0.98	−0.8

Left: 3.21, 3.0201, 3.002001

Right: $-0.8,\ -0.98,\ -0.998$

The two sides head to different values, so the table suggests the limit does not exist.

DNE (suggested)

$\lim_{x\to -1^-} f(x) \approx$ 3 $\lim_{x\to -1^+} f(x) \approx$ −1 $\lim_{x\to -1} f(x) \approx$ DNE

Evaluate each limit. (2 points each)

7. $\lim_{x\to -2^+} \dfrac{x^2+2x}{|x+2|}$

Substitution produces 0/0: indeterminate.

For $x > -2$, $|x+2| = x+2$.

$$\lim_{x\to -2^+} \frac{x(x+2)}{x+2}$$
$$= \lim_{x\to -2^+} x = -2$$

-2

8. $\lim_{x\to 4} \dfrac{x^2-16}{x^2-x-12}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 4} \frac{(x-4)(x+4)}{(x-4)(x+3)}$$
$$= \lim_{x\to 4} \frac{x+4}{x+3} = \frac{8}{7}$$

$\frac{8}{7}$

9. $\lim_{x\to 5} \dfrac{\sqrt{2x-1}-3}{x^2-25}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 5} \frac{(2x-1)-9}{(x^2-25)\left(\sqrt{2x-1}+3\right)}$$
$$= \lim_{x\to 5} \frac{2(x-5)}{(x-5)(x+5)\left(\sqrt{2x-1}+3\right)}$$
$$= \lim_{x\to 5} \frac{2}{(x+5)\left(\sqrt{2x-1}+3\right)}$$
$$= \frac{2}{(10)(3+3)} = \frac{2}{60} = \frac{1}{30}$$

$\frac{1}{30}$

10. $\lim_{x\to 1} \dfrac{\frac{x}{x+1}-\frac{1}{2}}{x-1}$

Substitution produces 0/0: indeterminate.

LCD: 2(x + 1)

$$\lim_{x\to 1} \frac{2x-(x+1)}{2(x+1)(x-1)}$$
$$= \lim_{x\to 1} \frac{x-1}{2(x+1)(x-1)}$$
$$= \lim_{x\to 1} \frac{1}{2(x+1)} = \frac{1}{2(2)} = \frac{1}{4}$$

$\frac{1}{4}$

11. For $f(x) = \sqrt{2x+1}$, find $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$

$$\lim_{h\to 0} \frac{\sqrt{2x+2h+1}-\sqrt{2x+1}}{h}$$

Multiply by the conjugate.

$$= \lim_{h\to 0} \frac{(2x+2h+1)-(2x+1)}{h\left(\sqrt{2x+2h+1}+\sqrt{2x+1}\right)}$$
$$= \lim_{h\to 0} \frac{2}{\sqrt{2x+2h+1}+\sqrt{2x+1}}$$
$$= \frac{2}{2\sqrt{2x+1}} = \frac{1}{\sqrt{2x+1}}$$

$\frac{1}{\sqrt{2x+1}}$